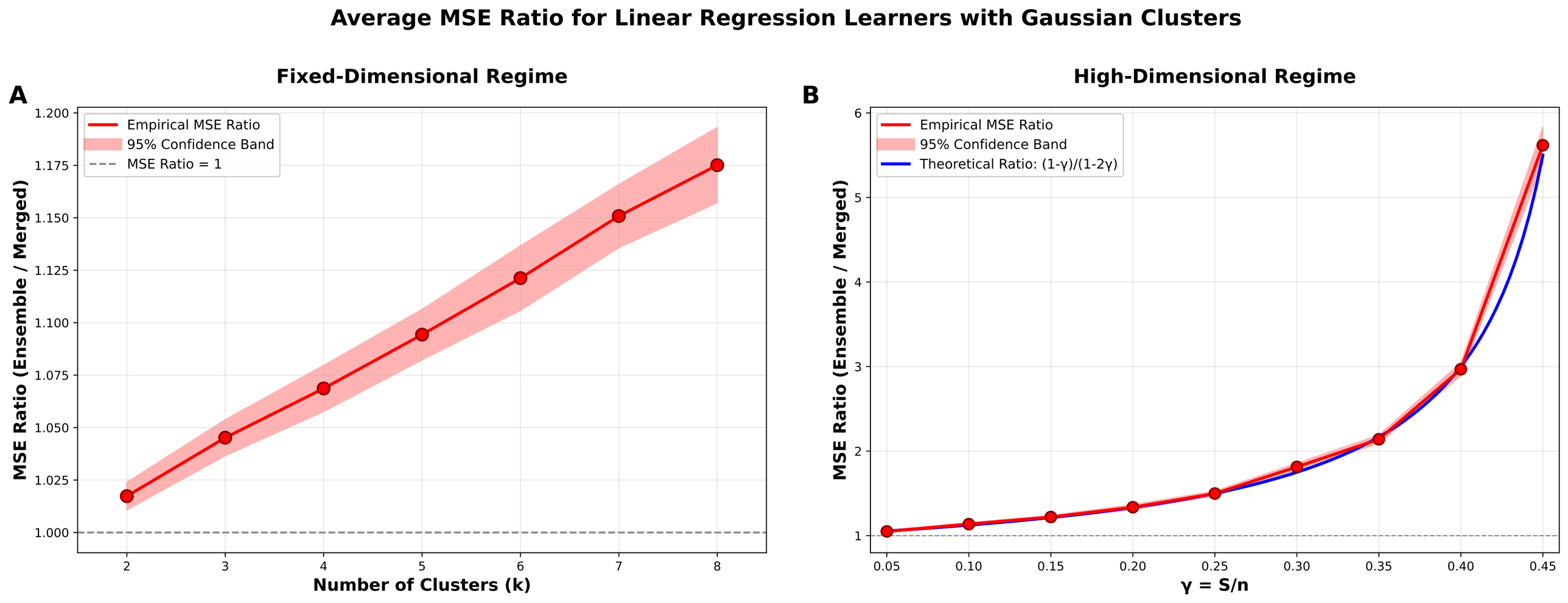
Average MSE Ratio for Linear Regression Learners with Gaussian Clusters
A
Fixed-Dimensional Regime
Empirical MSE Ratio
95% Confidence Band
MSE Ratio = 1
MSE Ratio (Ensemble / Merged)
1.200
1.175
1.150
1.125
1.100
1.075
1.050
1.025
1.000
2
3
4
5
6
7
8
Number of Clusters (k)
B
High-Dimensional Regime
Empirical MSE Ratio
95% Confidence Band
Theoretical Ratio: (1-γ)/(1-2γ)
MSE Ratio (Ensemble / Merged)
6
5
4
3
2
1
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45
γ = S/n